

t-test



Dr. Mohammad Rafiqul Islam
Associate Professor of Statistics
Dept. of Mathematics and Natural Sciences (MNS)
BRAC University, Dhaka, Bangladesh
mrafiq@bracu.ac.bd

t-test

- ❑ z-test and t-test are basically the same
 - ❑ where $n < 30$, the t-tests should be used
 - ❑ When variance (sd) is unknown t-test is used instead of z-test
 - ❑ the distributions should be normal for the equal and unequal variance t-test
 - ❑ For t-test all individuals must be selected at random from the population
 - ❑ t-test is sort form of Student's t test
- **Note that When variance (sd) is unknown but $n > 30$,t-test is preferred instead of z-test**

Hypothesis Tests When You Don't Know Your Population Parameters. Why should we prefer t-test?

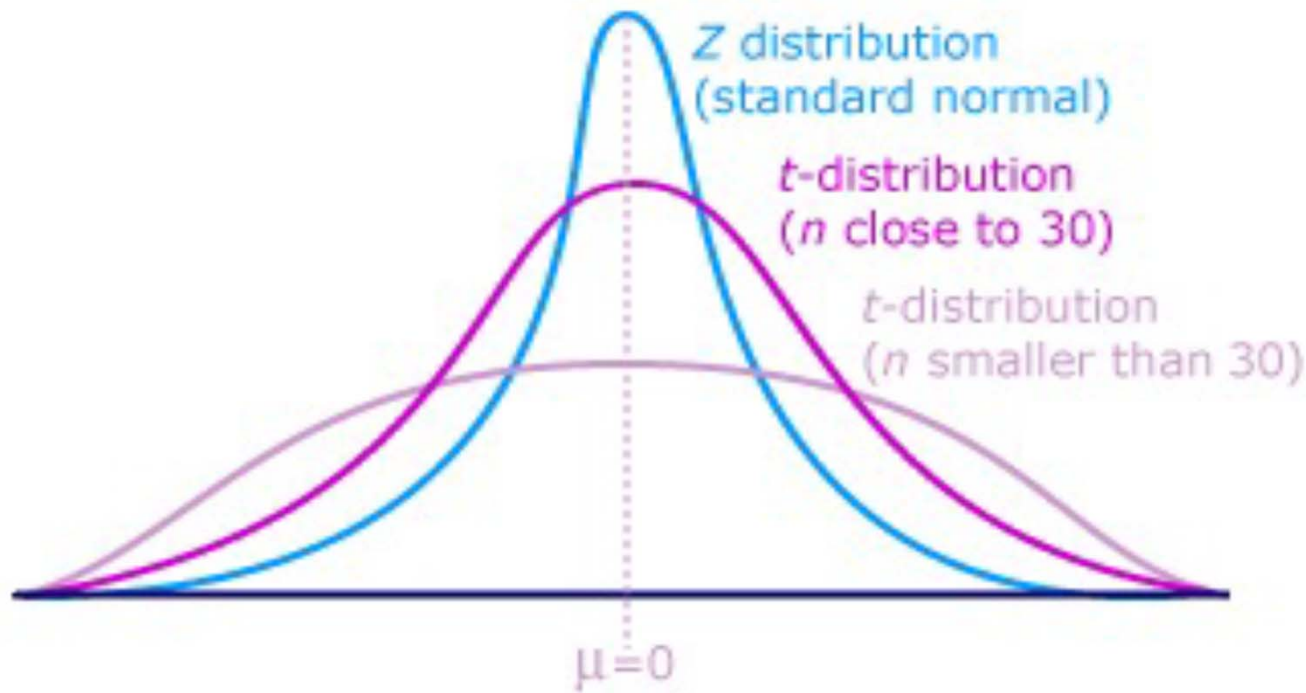
- A z-test makes for an easy hypothesis test, but most of the time we can't use it.
- The reality is that most of our analyses are done when we don't know what is true about a population.
- We don't know the true population mean, and we certainly don't know the true population standard deviation.
- Instead, we want to conduct a hypothesis test using only our sample and the information it can provide us.
- How can we do this?

How We Learned to Test Hypotheses on Samples, without Knowledge of the Population

- Back in the early 1900's, William Sealy Gosset who is working as a chemist at a brewing company (producing beer) in Ireland
- He was working with very small samples,
- He noticed that as his sample sizes changed, the shape of the distribution changed as well
- the distributions of the mean differed significantly from the normal distribution.
- He published his results under the pseudonym 'Student' and
- this concept and the distributions for small sample sizes are now known as "Student's t-distributions."

- The Student's t-distribution is similar to the normal distribution,
- except it is more spread out and wider in appearance, and has thicker tails.
- As the number of observations gets larger, the t-distribution shape becomes more and more like the shape of the normal distribution.
- In fact, if we had an infinite number of observations, the t distribution would perfectly match the normal distribution.
- It is the t-distribution that allows us to test hypotheses when we don't know the true population standard deviation, as we'll see below.

FIGURE



- As this statistical tool allows for testing of hypotheses with relatively small sample sizes.
- The t-test is very popular – millions of t-tests are performed every day in the market research industry alone

Conditions for Using t-test

The t-distribution can be used with any statistic having a bell-shaped distribution. the Central Limit Theorem predicts a roughly normal distribution under any of the following conditions:

- The population distribution is normal; or
- The sampling distribution is symmetric and the sample size is ' ≤ 15 ;' or
- The sampling distribution is moderately skewed and the sample size is $16 \leq n \leq 30$; or
- The sample size is greater than 30, without outliers.

The one-sample t-test.

Formula for one sample t test

In calculating the t-test statistic, we use the formula:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

T is the test statistic and has n-1 degrees of freedom.

$\bar{x}$ is the sample mean

μ is the population mean under the null hypothesis.

s is the sample standard deviation calculated from the sample

(if s is not given we have to calculate from the sample)

n is the sample size

$s/\sqrt{n}$ the estimated standard error

For Hypothesis Testing by t-test, you need to know about degrees of freedom that is generally one less than the number of data points, thus degrees of freedom(df) = n-1.

Degrees of freedom are essentially the number of samples

that have the 'freedom' to change without affecting the sample mean

The one-sample t-test.

Example 1

Duracell manufactures batteries and the CEO of this company claims will last an average of 300 hours under normal use. A researcher randomly selected 20 batteries from the production line and tested these batteries. The tested batteries had a mean life span of 270 hours with a standard deviation of 50 hours. Do we have enough evidence to suggest that the claim of an average lifetime of 300 hours is true?

Example(The one-sample t-test): solution

Step 1: Hypothesis Clearly state the Null and Alternative Hypothesis

$H_0 : \mu = 300$ (claim is true)

$H_A : \mu \neq 300$ (claim is false)

Step 2:Hypothesis : We'll use the standard significance level of 0.05,

Step 3: Compute the test statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Where , sample mean, $\bar{x} = 270$, population mean, $\mu = 300$, s =sample standard deviation=50, n =sample size=20

Now, we'll use that value in our t-test formula:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = (270 - 300)/(50/\sqrt{20}) = -2.68$$

As the test is two-tailed, we take the absolute value of t

¹⁰
 $|t| = |-2.68| = 2.68$

The one-sample t-test.

Example: solution

Step 4: Critical region: Degrees of freedom for this test is $df=n-1=(20-1)= 19$. With 19 df and at the 0.05 significance level tabulated value of t is 2.093. As the test is two-tailed the critical region is $CR \geq 2.093$.

Step 5: Decision

Since the calculated value of $|t|$, $2.68 > 2.093$ falls in the critical region, we reject the null hypothesis and accept the alternative. Thus, the claim of the CEO is not true that a battery last on an average of 300 hours

P-value approach: For two-tailed

Example: solution: The one-sample t-test.

Step 4: P-value: Find the probability that t is beyond (more extreme than) your test statistic

p value of test statistics $t=2.68$ with $20-1=19$ degrees of freedom

$P\text{-value} = 2 * P(t > 2.68) = 2 * [1 - P(t < 2.68)] = 2 * [1 - 0.992593] = 2 * [0.00741] = 0.01482$

[in excel, =T.DIST(2.68,19,TRUE)=0.992593]

[in R programming: >1-pt(abs(t),df=n-1)=1-pt(2.68,19)=1-0.9925929=0.007407105,

then $P\text{-value} = 2 * 0.007407105 = 0.01481421$]

As $p\text{-value} < 0.05$, it is unlikely that we will get better result than our test statistics of 2.68 which is favor of alternative hypothesis H_A , for the given sample.

Step 5: Statistical decision:

we know that if $p\text{-value} < 0.05$, we reject the null hypothesis at 5% level of significance

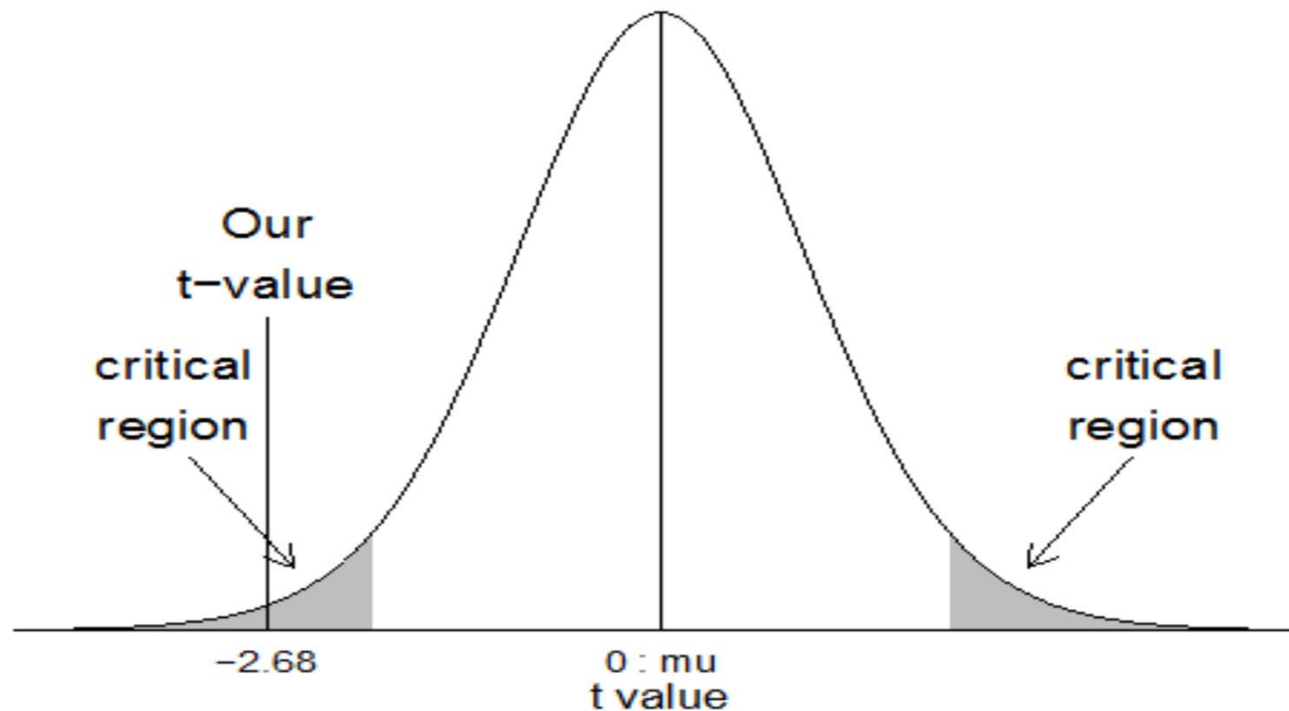
So, the $p\text{-value} 0.01482 < 0.05$, we reject the null hypothesis at 5% level of significance.

Thus, the claim of the CEO is not true that a battery last on an average of 300 hours

Statistical Hypothesis Testing

t-test

The one-sample t-test. Let us see visually



Hypothesis Step 5: Interpret your results

Since our calculated t-test value is outside of our t-critical value –it lies in the critical region –we reject the Null Hypothesis. The average battery life of the sample is significantly different from the average battery life claim by the CEO.

The one-sample t-test.

Example 2: Building specifications in a certain city require that the average breaking strength of sewer pipe be more than 2500 kilograms per foot of length. The mean breaking strength for a sample of 18 sections of sewer pipe was found to be average 2550 kilograms per foot with standard deviation is 120 kilograms per foot. Does enough evidence exist to conclude at the level of significance of 5% that the pipe meet specification?
[remember that our alternative hypothesis will be desired result that is what do we want]

The one-sample t-test.

Step 1: State the hypotheses.

$H_0: \mu = 2500$ (or $\mu \leq 2500$) (the pipe does not meet specification)

$H_A: \mu > 2500$ (the pipe meets specification)

Step 2: Specification of the level of significance, $\alpha = 5\% = 0.05$.

Step 3: Compute the t test statistic z as

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Here sample mean, $\bar{x} = 2550$, population mean, $\mu = 2500$, s.d = 120, $n = 18$
 $t = (2550 - 2500) / (120 / \sqrt{18}) = 1.77$

Step 4: Calculation of the tabulated value: this is one-tailed test

With $n - 1 = 18 - 1 = 17$ degree of freedom (df) the *tabulated value of t at 5% level of significance is 1.740*. So the for the one-tailed right hand test critical region is

$C.R > 1.740$

Step 5: Statistical decision:

Since calculated value of t $1.77 > 1.740$ falls in the critical region, we reject the null hypothesis **and thus we accept the alternative**. We conclude that the pipe meets specification.

The one-sample t-test.

Example: solution: The one-sample t-test.

P-value approach: For one tailed

Step 4: P-value: the probability that t is beyond (more extreme than) your test statistic for $H_A: \mu > 2500$ will be $p(t > 1.77)$

p value of test statistics $t=1.77$ with $18-1=17$ degrees of freedom

P -value = $*P(t > 1.77) = 1 - P(t < 1.77) = 1 - 0.952671 = 0.047329$

[in excel, =T.DIST(1.77,17,TRUE)=0. 0.952671]

[in R programming: $>1-pt(t,df=n-1)=1-pt(1.77,17)=1-0. 0.9526707=0.0473293$]

Step 5: Statistical decision:

we know that if $p\text{-value} < 0.05$, we reject the null hypothesis at 5% level of significance

So, the $p\text{-value } 0.047329 < 0.05$, we reject the null hypothesis at 5% level of significance. Thus, We conclude that the pipe meets specification.

The one-sample t-test.

Example 3

You have just taken ownership of a pizza shop. The previous owner told you that you would save money if you bought the mozzarella cheese in a 4.5 pound slab. Each time you purchase a slab of cheese, you weigh it to ensure that you are receiving 72 ounces of cheese. The results of 7 random measurements are 70, 69, 73, 68, 71, 69 and 71 ounces. Are these differences due to chance or is the distributor giving you less cheese than you deserve?

Example 4

The Chairman of a dept. is concerned that the student's grade point averages have changed dramatically in recent years. The graduating seniors' mean GPA over the last five years is 2.75. The dean randomly samples 256 seniors from the last graduating class and finds that their mean GPA is 2.85, with a sample standard deviation of 0.65.

- What would the null and alternative hypotheses be for this scenario?
- Test the null hypothesis and explain your decision

Example 5: Aircrew escape systems are powered by a solid propellant. The burning rate of this propellant is an important product characteristic. Specifications require that mean burning rate must be 50 centimeters per second. The experimenter selects a random sample of 25 propellants and obtains a sample average burning rate of 48.7 centimeters per second with standard deviation of 2 centimeters per second. What conclusion can be drawn from the sample

The one-sample t-test.

Example 6: A health-care actuary has been investigating the cost of maintaining the cancer patients within its plan. These people have typically been running up costs at the rate of \$1,240 per month. A sample of 15 cases for November (the first 15 for which complete records were available) and an average cost of \$1,080, with a standard deviation of \$180. Is there any evidence of a significant change?

Example 7. The hourly French fried potato output by the Krisp-o-Matic fry machine is advertised to be 150 pounds. For the new machine purchased by the Burger Heaven drive-in, tests were run for 22 different one-hour periods, producing an average production of 143 pounds, with a standard deviation of 17 pounds. At the 5% level of significance, does the Burger Heaven management have grounds for complaints?

Table T Critical Values of the *t* Distribution

<i>df</i>	One-Tail = .4 Two-Tail = .8	.25 .5	.1 .2	.05 .1	.025 .05	.01 .02	.005 .01	.0025 .005	.001 .002	.0005 .001
1	0.325	1.000	3.078	6.314	12.706	31.821	63.657	127.32	318.31	636.62
2	0.289	0.816	1.886	2.920	4.303	6.965	9.925	14.089	22.327	31.598
3	0.277	0.765	1.638	2.353	3.182	4.541	5.841	7.453	10.214	12.924
4	0.271	0.741	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610
5	0.267	0.727	1.476	2.015	2.571	3.365	4.032	4.773	5.893	6.869
6	0.265	0.718	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959
7	0.263	0.711	1.415	1.895	2.365	2.998	3.499	4.029	4.785	5.408
8	0.262	0.706	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041
9	0.261	0.703	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.781
10	0.260	0.700	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.587
11	0.260	0.697	1.363	1.796	2.201	2.718	3.106	3.497	4.025	4.437
12	0.259	0.695	1.356	1.782	2.179	2.681	3.055	3.428	3.930	4.318
13	0.259	0.694	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.221
14	0.258	0.692	1.345	1.761	2.145	2.624	2.977	3.326	3.787	4.140
15	0.258	0.691	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.073
16	0.258	0.690	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.015
17	0.257	0.689	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.965
18	0.257	0.688	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.922
19	0.257	0.688	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.883
20	0.257	0.687	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.850
21	0.257	0.686	1.323	1.721	2.080	2.518	2.831	3.135	3.527	3.819
22	0.256	0.686	1.321	1.717	2.074	2.508	2.819	3.119	3.505	3.792
23	0.256	0.685	1.319	1.714	2.069	2.500	2.807	3.104	3.485	3.767
24	0.256	0.685	1.318	1.711	2.064	2.492	2.797	3.091	3.467	3.745
25	0.256	0.684	1.316	1.708	2.060	2.485	2.787	3.078	3.450	3.725
26	0.256	0.684	1.315	1.706	2.056	2.479	2.779	3.067	3.435	3.707
27	0.256	0.684	1.314	1.703	2.052	2.473	2.771	3.057	3.421	3.690
28	0.256	0.683	1.313	1.701	2.048	2.467	2.763	3.047	3.408	3.674
29	0.256	0.683	1.311	1.699	2.045	2.462	2.756	3.038	3.396	3.659
30	0.256	0.683	1.310	1.697	2.042	2.457	2.750	3.030	3.385	3.646
40	0.255	0.681	1.303	1.684	2.021	2.423	2.704	2.971	3.307	3.551
60	0.254	0.679	1.296	1.671	2.000	2.390	2.660	2.915	3.232	3.460
120	0.254	0.677	1.289	1.658	1.980	2.358	2.617	2.860	3.160	3.373
∞	0.253	0.674	1.282	1.645	1.960	2.326	2.576	2.807	3.090	3.291

Source: From *Biometrika Tables for Statisticians*, Vol. 1, Third Edition, edited by E. S. Pearson and H. O. Hartley, 1966, p. 146.
 Reprinted by permission of the Biometrika Trustees.